

COSC 3P32 – File and Database Systems

Winter 2026

Assignment #3

Due Date: 25th March, 2026, 11:59pm

No late assignments are accepted

This assignment accounts for 5% of your final grade and is worth a total of 50 marks.

This assignment is to be completed *individually*.

Question 1 [4 marks]

Consider the following relation:

Assignments(*cid*, *semester*, *aid*, *type*, *duedate*, *weight*, *maxmarks*).

In this relation, *cid* is the course ID and *aid* is the assignment ID.

Using a single letter for each attribute, represent *cid* as C, *semester* as S, *aid* as A, *type* as T, *duedate* as D, *weight* as W, and *maxmarks* as M.

Express each of the following statements as functional dependencies (FDs):

- The key of Assignments is (*cid*, *semester*, *aid*).
- There is only one assignment for a given course, semester, and due date combination.
- All assignments of the same weight for a given course and semester have the same max. marks.
- All assignments of the same type for the same course have the same weight.

Question 2 [20 marks]

Consider the relation schema $R = VWXYZ$,

with the set of FDs $F = \{VW \rightarrow XY, V \rightarrow X, W \rightarrow Z, YZ \rightarrow Z, Z \rightarrow V\}$.

- [3x2 marks] Give three **independent** reasons why the following table cannot possibly be a legal instance of R.

V	W	X	Y	Z
v1	w1	x1	y1	z1
v2	w1	x2	y2	z1
v2	w2	x1	y2	z2

- [2x2 marks] Show the FDs that hold on each of $R_1 = VWX$ and $R_2 = WYZ$. You should not show trivial FDs, or those implied by others in the same set, but you must show all others.
- [2 marks] Is the decomposition of R into $R_1 = VWX$ and $R_2 = WYZ$ lossless-join? Why or why not?
- [5 marks] Find a lossless-join BCNF decomposition of R (note that you *cannot* use the decomposition from part (c)). At each step, you must show all FDs that hold on each of the resulting relations. You should not need to show FDs that are trivial or implied by other FDs in the same set, but you must show all others.
- [3 marks] Is your decomposition found in part (d) dependency-preserving? Why or why not?

Question 3 [11 marks]

Consider the relation schema $R=ABCDEFGH$,
with the set of FDs $F=\{B \rightarrow AGH, CD \rightarrow B, C \rightarrow EG, E \rightarrow AD, G \rightarrow E\}$.

- a) [6 marks] Find a minimal cover for F .
- b) [2 marks] Find all candidate key(s) of R .
- c) [3 marks] Use 3NF Synthesis to find a 3NF decomposition of R that is both lossless-join and dependency-preserving. You must show all FDs that hold on each of the resulting relations. You do not need to show FDs that are trivial or implied by other FDs in the same set, but you must show all others.

Question 4 [15 marks]

Consider the following database schema, in which the keys of each relation are underlined:

Restaurant(*rname*, *address*, *phone*, *stars*)

Chef(*cname*, *specialdish*, *rating*)

CooksFor(*cname*, *rname*, *salary*)

Offers(*rname*, *dishname*, *price*)

For each of the following general constraints, create CHECK constraints, assertions or triggers (as appropriate) to ensure that the constraint holds. Each of these is to be considered independently, i.e. you do not need to worry about any of them affecting each other.

Note: in the case of CHECK constraints, you do *not* need to provide the entire set of statements required to create the appropriate table, just the CHECK constraint itself.

- a) [2 marks] Write a CHECK constraint to ensure that every chef has a rating between 0 and 10.
- b) [2 marks] Write a CHECK constraint to ensure that no chef salary for restaurants called *BigMac* is more than 100.
- c) [3 marks] Write an assertion to ensure that every restaurant must have at least one chef.
- d) [3 marks] Write an assertion to ensure that for every dish, there is at least one chef for whom that is their special dish.
- e) [5 marks] When the number of stars assigned to a restaurant increases, then all chefs who cook for that restaurant must receive a 10% raise. Write a trigger that will modify *salary* accordingly, using the SQL:1999 standard.

Submission Requirements:

1. Ensure that you have fully complied with all notation and formatting requirements for all questions.
2. All submissions must be made via Brightspace.
3. Your submission must consist of a single PDF file containing all answers to all questions.